



HANOVER & TYKE HOTEL: KEY PERFORMANCE INSIGHTS ★

POWER BI ANALYSIS



Home



OverView



Sales & Business Insights



Customer Analysis



Cancellation Analysis



Hotel Performance





Project Overview

Business Challenge

Understand booking patterns, identify operational inefficiencies, and uncover growth opportunities across hotel properties

Objectives

Analyze customer behavior, diagnose cancellation drivers, optimize revenue channels, and support strategic decision-making

Tools Used

SQL Server for data extraction & preparation, Power BI for modeling, DAX measures, analysis & visualization

Data Cleaning & Normalization



Null Handling

Replaced all null values in the children column with 0

Agent ID Correction

missing values in the agent column were replaced with **Agent ID 22** to ensure data completeness

Data Type Corrections

agent and company columns converted from numeric to text — they are identifiers, not values

Outlier Detection

Corrected ADR outlier: **5,400** → **540** for Daniel Walter booking

Quality Assurance

Verified all remaining nulls and inconsistencies — dataset confirmed clean and ready

Data Model: Star Schema

Why Star Schema?

Optimizes query performance, simplifies analysis, and scales efficiently for complex reporting.

Fact Table

HotelBooking — the core transactional table at the center of the model

Relationships

One-to-Many from all dimension tables into the HotelBooking fact table

Dimension Tables

Date

Country

Hotels

Meal

MarketSegment

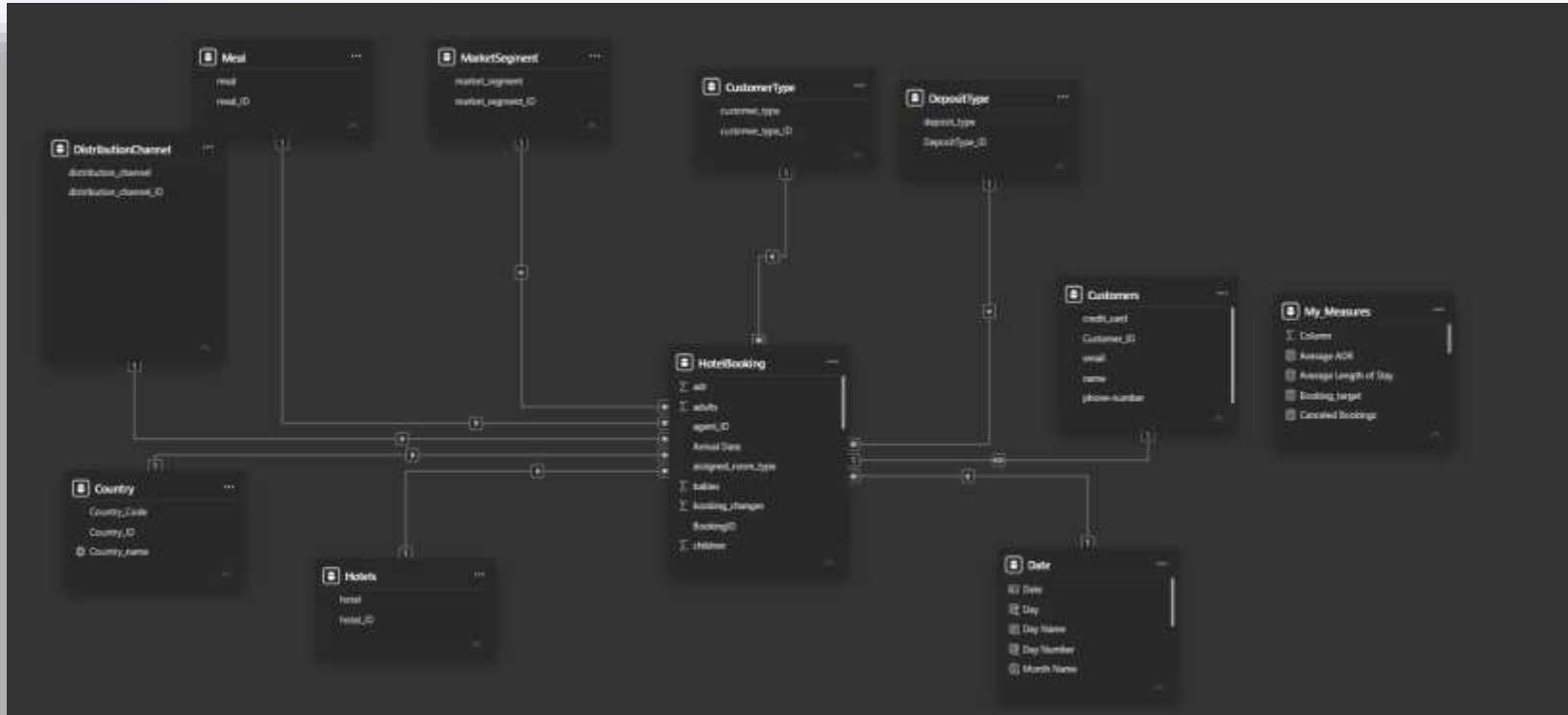
DistributionChannel

CustomerType

DepositType

Customers

Data Model: Star Schema



Key DAX Measures



Total Revenue

Sum of ADR × total nights
across all bookings



Cancellation Rate

Canceled bookings ÷ total
bookings — tracks revenue
leakage



Avg. Length of Stay

Total nights ÷ total bookings —
measures guest engagement



Repeat Guest %

Repeat bookings ÷ total bookings
— indicates customer retention
health



Checked Out

Bookings with "Check-Out"
status — actual revenue-
generating stays

CHAPTER

Dashboard Insights

From raw data to visual intelligence — exploring what the numbers reveal





Business Snapshot

Real time view of revenue, ADR, Booking trends and Cancellation rate

hotel

All

Country

All

Year

All



\$42.72M

Total Revenue



\$101.79

Avg. ADR



3

Avg. Stay days



37%

Cancellation Rate



3.2%

Repeat Guest %

Booking Trends



Total Bookings

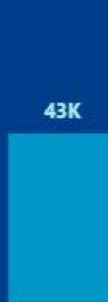


75K



Check-Out

43K



Canceled

1K

No-Show

Business Snapshot: Key Findings & Recommendations

Key Findings

Revenue & Volume

\$42.72M total revenue

\$101.79 ADR

3-day average stay

119.39K total bookings

⚠️ Critical — Cancellations

37% cancellation rate —
43K canceled vs. 75K
check-outs = massive
revenue leakage

Retention Crisis

Only **3.2% repeat guests**
— business relies heavily
on constant new customer
acquisition

Seasonality

Strong summer peak (**Aug:**
13.9K bookings) with sharp
winter decline (**Nov–Dec:**
6.8K)

Recommendations



Launch a Loyalty Program

Tiered rewards or direct-
booking discounts to
raise lifetime value and
lower acquisition costs



Smooth Seasonal Dips

Winter campaigns,
corporate retreat
packages, and holiday
specials targeting the
Nov–Jan slump



HANOVER & TYKE HOTEL

Sales & Business Insights

hotel

All

Country

All

Year

All

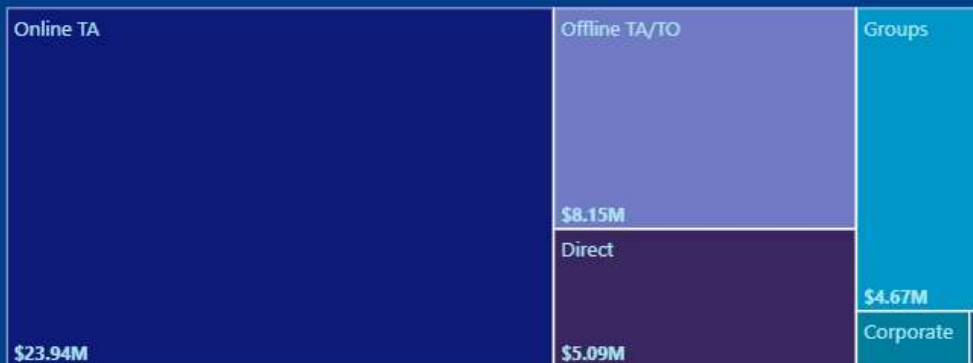
Total Bookings by hotel



Total Revenue by Month



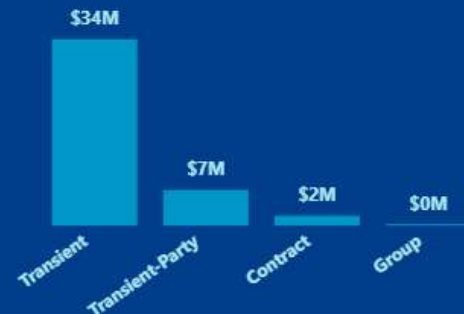
Total Revenue by market_segment



Total Revenue by hotel



Total Revenue by customer_type



Sales & Business Insights: Key Findings & Recommendations

What We Found

Hotel Comparison

City Hotel drives **79K bookings** and **\$25M revenue** vs. Resort Hotel at 40K bookings and \$17M revenue

Distribution Channel Imbalance

OTAs generate **\$23.94M** while Direct bookings lag at only **\$5.09M**

Revenue Seasonality

Revenue peaks in **August (\$7.9M)** and drops significantly in late fall/winter

What We Recommend

Shift to Direct Bookings

Launch "**Book Direct & Save**" initiative with free breakfast, late checkout, and 5% discount to reduce OTA commission costs (typically 15–25%)

Optimize City Hotel Pricing

Implement **dynamic pricing** during peak corporate seasons to push ADR higher on the property driving 59% of total revenue

Customer Demographics & Behavior

hotel

All

Country

All

Year

All

222K

Total Adults



68K

Special Requests



12.40K

Total Children



Customer Types Distribution

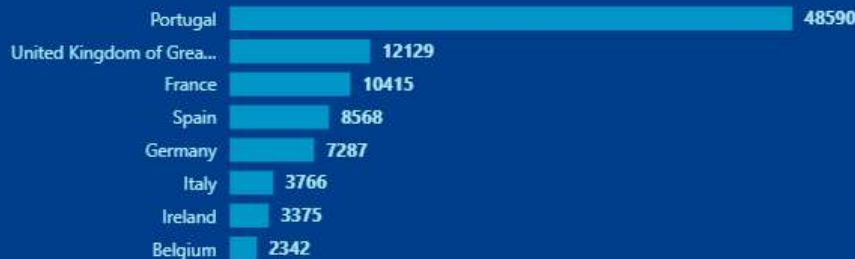
4076 (3.41%)
25124 (21.04%)

89613 (75.06%)

customer_type

- Transient
- Transient-Party
- Contract
- Group

Total Bookings by Country



Total Bookings by repeated guests

4K (3.19%)



116K (96.81%)

is_repeated_guest

- 0
- 1

Guest Composition (Adults vs Children vs Babies).

Total Children Total Babies Total Adults



Customer Demographics: Key Findings

48.6K

Portugal Bookings

Dominant origin market — 4× the volume of the UK (12,129 bookings)

222K

Adult Guests

Guest composition is almost entirely adults — children (12.4K) and babies (949) are minimal

75%

Transient Travelers

Transient (75.06%) + Transient-Party (21.04%) dominate — virtually no group or contract business

- Geographic and customer-type concentration signals both a dependency risk and a targeted growth opportunity in underpenetrated segments.

Strategic Recommendations

Targeted Marketing

Double down on localized marketing in **Portugal, the UK, and France**. Ensure the website and booking engines are **natively translated and optimized** for these regions.

Refine the Hotel's Positioning

Since the demographic is heavily **adult and transient**, shift amenities and marketing to appeal to **couples, solo travelers, and business professionals** rather than investing heavily in **family or children's entertainment**.



Cancellation Analysis

hotel

All

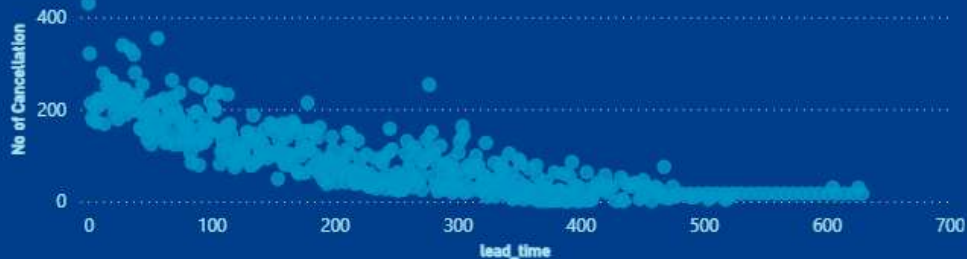
Country

All

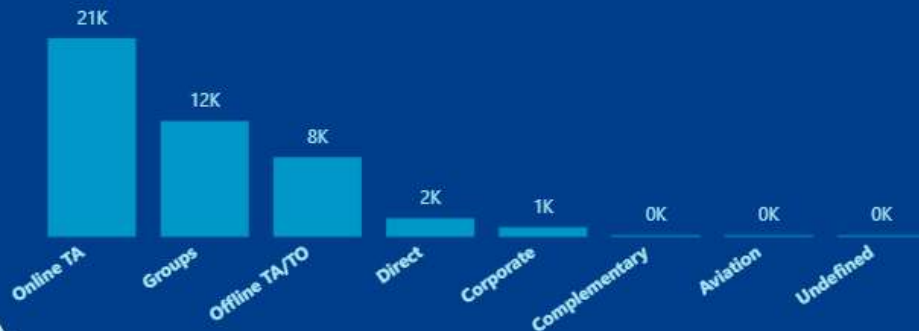
Year

All

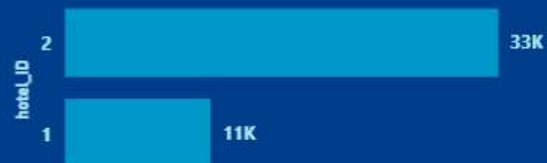
cancellation by lead_time



cancellation by market_segment



cancellation by hotel



cancellation by deposit type



Cancellation Analysis: Key Findings & Recommendations

What We Found

- **Deposit Type Risk:** 67.14% of cancellations come from "No Deposit" bookings; 32.77% from "Non Refund" bookings
- **Lead Time Pattern:** Massive cluster of cancellations occurs when lead time is under 200 days
- **Segment Risk:** Online TAs account for 21K cancellations, followed by Groups at 12K
- **Property Discrepancy:** Hotel 2 experiences **3× the cancellation volume** (33K) compared to Hotel 1 (11K)

What We Recommend

Overhaul Deposit Policies

Require first-night deposit for OTA bookings to guarantee intent and reduce ghost bookings

Investigate "Non-Refund" Cancellations

Clarify revenue retention on 14.49K "Non Refund" cancellations; renegotiate OTA contracts if needed

Data-Driven Overbooking

Use predictable OTA and Group cancellation patterns to set overbooking limits and maximize occupancy



HANOVER & TYKE HOTEL

Hotel Performance

hotel

All

Country

All

Year

All

agent_ID	Total Bookings	Confirmed Bookings	Canceled Bookings
9	31961	19070	13264
22	16722	12855	4075
240	13922	8539	5484
1	7191	1911	5280
14	3640	3074	652
7	3539	3093	474
6	3290	2273	1025
250	2870	2389	513
241	1721	1492	236
28	1666	1613	110
8	1514	1127	427
3	1336	567	771
37	1230	523	717
19	1061	282	780
40	1039	962	84
314	927	772	163
21	875	369	506
229	786	303	484
242	780	535	260
83	696	516	189
29	683	138	546
171	607	435	172
Total	119390	76373	44224

Total Bookings by market_segment

market_segment

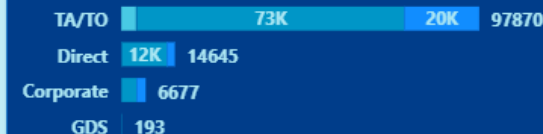


Total Bookings



Distribution Channels

customer_type ● Contract ● Group ● Transient ● Transient-Party



Total Bookings by meal and hotel

hotel ● City Hotel ● Resort Hotel



Hotel Performance: Key Findings & Recommendations

What We Found

Agent Performance Paradox

Agent ID 9 brings massive volume (**31,961 bookings**) but carries exceptionally high cancellations (**13,264**)

Meal Plan Dominance

Bed & Breakfast (BB) is the clear favorite — 62K+ at City Hotel, 30K+ at Resort; Half Board and Self Catering rarely utilized

Funnel Drop-off

37% of top-of-funnel bookings are canceled, leaving check-ins at only **63%** of initial activity

What We Recommend

□ Agent Accountability

Audit quality of traffic from **Agent 9 and Agent 22**; shift commission structures to reward **completed stays** rather than raw booking volume

□ Upsell Meal Plans

Create automated pre-arrival emails offering discounted **Half Board upgrades** to capture additional food & beverage revenue

Summary & Next Steps

Hanover & Tyke Hotel generates strong top-line revenue (\$42.72M) and healthy booking volumes, but profitability and efficiency are severely impacted by a 37% cancellation rate, over-reliance on OTAs, and critically low customer retention (3.2% repeat guests).

Key Takeaways

- Fix the cancellation crisis through deposit policy overhauls and OTA renegotiation
- Shift revenue mix from OTAs to Direct bookings to improve margins
- Build customer loyalty through retention programs to reduce acquisition costs
- Optimize pricing and operations by property and season

Next Steps

01

Loyalty Program

Implement loyalty program and "Book Direct & Save" initiative

02

Agent Restructure

Audit and restructure agent commission models

03

Seasonal Marketing

Launch seasonal marketing campaigns for winter occupancy

04

OTA Renegotiation

Renegotiate OTA contracts with deposit and refund policy changes

THANK YOU